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Assignment - Parameter Estimation

Q1:- Let $(x_1, x_2 -- x_n)$ be a random sample of size n taken from a Normal population with parameters: mean $= \theta_1$ and variance θ_2. Find the maximum likelihood estimator of these two parameter.

given $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x_i-\mu)^2}{2\sigma^2}}$

Sol:- For a random variable x_i,

$$f(x_i) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}} \quad -- \quad (1)$$

As $x_1, x_2 -- x_n$ are independent, Likelihood function

$$L = \prod_{i=1}^{n} f(x_i)$$

$$= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x_i-\mu)^2}{2\sigma^2}}$$

$$\boxed{L = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \prod_{i=1}^{n} e^{\frac{-(x_i-\mu)^2}{2\sigma^2}}} \quad --- \quad (2)$$

Take log on both sides,

$$\ln L = \ln\left(\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \prod_{i=1}^{n} e^{\frac{-(x_i-\mu)^2}{2\sigma^2}}\right)$$

$$= \ln\left(\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n\right) + \ln\left(\prod_{i=1}^{n} e^{\frac{-(x_i-\mu)^2}{2\sigma^2}}\right)$$

$$= n \ln\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) + \sum_{i=1}^{n} \ln\left(e^{\frac{-(x_i-\mu)^2}{2\sigma^2}}\right)$$

$$= n \ln\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \sum_{i=1}^{n} \frac{(x_i-\mu)^2}{2\sigma^2}$$

$$\boxed{\ln L = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2} \quad -- \quad (3)$$

Differentiate it with respect to μ,

$$\frac{\partial \log L}{\partial \mu} = -\frac{1}{2\sigma^2}\frac{\partial}{\partial \mu}\sum_{i=1}^{n}(x_i-\mu)^2$$

$$= -\frac{1}{2\sigma^2}(2)\sum_{i=1}^{n}(x_i-\mu)(-1)$$

$$\boxed{\frac{\partial \log L}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu)} \quad ---(4)$$

Diff wrt to σ^2

$$\frac{\partial \log L}{\partial \sigma^2} = \frac{\partial}{\partial \mu}\left(-\frac{n}{2}\log\sigma^2\right) + \frac{\partial}{\partial \mu}\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right)$$

$$\boxed{\frac{\partial \log L}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(x_i-\mu)^2} \quad ---(5)$$

Equate eqn (4) and eqn (5) to 0;

$$\frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu) = 0$$

$$\sum_{i=1}^{n}(x_i-\mu) = 0$$

$$\sum_{i=1}^{n}x_i - \sum_{i=1}^{n}\mu = 0$$

$$\sum_{i=1}^{n}x_i - n\mu = 0$$

$$\boxed{\mu = \frac{\sum_{i=1}^{n}x_i}{n}} \quad ---(6).$$

$$-\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i-\mu)^2 = 0$$

$$-\frac{n}{2} + \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2 = 0$$

$$-n\sigma^2 + \sum_{i=1}^{n} (x_i-\mu)^2 = 0$$

$$n\sigma^2 = \sum_{i=1}^{n} (x_i-\mu)^2$$

$$\boxed{\sigma^2 = \frac{\sum_{i=1}^{n} (x_i-\mu)^2}{n}} \quad \text{--- (7)}$$

Ans:- $\theta_1(\text{Mean}) = \dfrac{\sum_{i=1}^{n} x_i}{n}$; $\theta_2(\text{Variance}) = \dfrac{1}{n}\sum_{i=1}^{n} (x_i-\mu)^2$

Q2:- Let $X_1, X_2 \text{---} X_n$ be a random sample from $B(m,\theta)$ distribution, where $\theta \in (0,1)$ is unknown and 'm' is a known positive number. Compute value θ using the M.L.E.

Sol:- for a random variable x_i;

pmf $\Rightarrow P(X_i = x_i) = {}^mC_{x_i}\, \theta^{x_i} (1-\theta)^{m-x_i}$ — ①

Likelihood function will be joint distribution function of $X_1, X_2 \text{---} X_n$

$$L = \prod_{i=1}^{n} {}^mC_{x_i}\, \theta^{x_i} (1-\theta)^{m-x_i}$$

$$L = \left(\prod_{i=1}^{n} {}^mC_{x_i}\right) \theta^{\sum x_i} (1-\theta)^{\sum(m-x_i)} \quad \text{— ②}$$

Taking log on both sides,

$$\ln L = \ln\left(\prod_{i=1}^{n} {}^mC_{x_i}\, \theta^{\sum x_i} (1-\theta)^{nm-\sum x_i}\right)$$

$$\boxed{\ln L = \ln\left(\prod_{i=1}^{n} {}^mC_{x_i}\right) \left(\sum x_i \log\theta\right) \left(nm - \sum x_i\right)\left(\log(1-\theta)\right)}$$

— ③

Taking derivative wrt θ

$$\frac{\partial \log L}{\partial \theta} = \frac{\partial}{\partial \theta}\left(\sum x_i \log\theta + (nm - \sum x_i)\log(1-\theta)\right)$$

$$= \sum x_i \frac{\partial \log\theta}{\partial\theta} + \frac{(nm - \sum x_i)\ \partial \log(1-\theta)}{\partial\theta}$$

$$\Rightarrow \frac{\sum x_i}{\theta} + \frac{(nm - \sum x_i)}{1-\theta}(-1)$$

$$\Rightarrow \frac{\sum x_i}{\theta} + \frac{nm - \sum x_i}{(\theta - 1)}$$

Equating it to zero;

$$\frac{\sum x_i}{\theta} = \frac{nm - \sum x_i}{(1-\theta)}$$

$$\frac{1-\theta}{\theta} = \frac{nm - \sum x_i}{\sum x_i}$$

$$1/\theta \cancel{-1} = \frac{nm}{\sum x_i} \cancel{-1}$$

$$\boxed{\theta = \frac{\sum x_i}{nm}}$$

Answer:- $\boxed{\theta = \frac{\sum_{i=1}^{m} x_i}{nm}}$